

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

Unit 2 assessment

Unit summary

About 97 per cent of Earth's water is salt, and most fresh water is ice and groundwater, so water security is a match between need and usable supply, not a count of litres. Pakistan's resources centre on the Indus system, already highly stressed, plus groundwater that can be abundant yet unsafe, as Thar's Chachro samples showed. Water pollution runs source, pathway, receptor, effect, and its types, pathogens, organic matter, nutrients, heavy metals, heat and silt, each match a source, an impact and a remedy at one link. Industrial effluents are treated in stages, primary for solids, secondary for organics, tertiary for metals and colour, and can then be recycled. Municipal solid waste runs through five stages ordered by the hierarchy, reduce, reuse, recycle, recover, dispose, with segregation at source deciding everything downstream. Hazardous hospital and industrial waste never enters the municipal stream: colour-coded segregation at generation, secure storage, licensed treatment.

Summative assessment

Multiple-choice questions (MCQs)

1. Of every 100 units of water on Earth, roughly how many are salt water?
(Remember) a) about half b) about 97 c) about 70 d) about 30
2. Water security is best defined as: (Understand) a) high annual rainfall b) reliable access to water of acceptable quality and quantity c) the number of large dams a country owns d) the price of a tanker of water
3. The Chachro study of Thar's groundwater found that: (Apply) a) the water was abundant and safe b) parameters including total dissolved solids

exceeded WHO drinking guidelines c) no measurements could be taken d) the desert depends only on rain

4. Which pollutant type causes eutrophication? (*Remember*) a) pathogens b) nutrients c) heat d) grit
5. Secondary treatment of effluent works mainly by: (*Understand*) a) settling grit in a tank b) bacteria consuming dissolved organic matter c) adsorbing metals onto charcoal d) filtering out colour
6. A tannery discharges chromium. Properly handled, the chromium is removed by: (*Apply*) a) primary settling alone b) dilution with canal water c) tertiary treatment built for metals d) discharging at night
7. The waste hierarchy, best to worst, runs: (*Remember*) a) recycle, reduce, reuse, dispose, recover b) reduce, reuse, recycle, recover, dispose c) dispose, recover, recycle, reuse, reduce d) reuse, reduce, recover, dispose, recycle
8. Segregation at source matters most because: (*Understand*) a) it makes the street look tidy b) mixed waste cannot be cleanly recycled or composted c) collection trucks prefer light bins d) it is required only of hospitals
9. In the burning-heap example, the most serious failure was that: (*Apply*) a) paper was being burned b) clinical waste had entered the municipal stream c) plastic bags were visible d) the heap stood near a school gate
10. Most of Earth's fresh water is held: (*Remember*) a) in rivers and lakes b) in ice caps, glaciers and groundwater c) in clouds d) in city storage tanks

Restricted-response questions (RRQs)

1. In two sentences, explain how a district can flood and still face drinking-water scarcity. (*Analyze*)
2. Name Pakistan's four main water resources and one challenge to each. (*Understand*)
3. Which half of water security, quantity or quality, did the Chachro study show failing? Give the finding and your reason. (*Apply*)

4. Name the three links of the pollution chain and one remedy at each link. *(Understand)*
5. In one sentence each, state what primary, secondary and tertiary treatment remove. *(Understand)*
6. Why is "the water looks clear" not evidence that a well is safe? Use two pollutant types in your answer. *(Analyze)*
7. Explain how a factory recycling its treated effluent serves both the factory and the river. *(Analyze)*
8. Why does reduce sit above recycle in the waste hierarchy? *(Understand)*
9. List three hospital risk-waste categories and state the handling rule for sharps. *(Understand)*
10. A city can fund a truck fleet or a segregation campaign with bins. Argue for one in three sentences, using the hierarchy. *(Analyze)*

Extended-response questions (ERQs)

1. Using the 100-droplet picture and the definition of water security, explain to a Class 8 pupil how their district could flood one year and face shortage the next. Name one household action and one city action that would reduce the shortage. *(Analyze)*
2. A settlement's open drain receives household grey water and a textile unit's discharge, and a hand pump stands nearby. Analyse the chain, the pollutant types and the receptors, and propose one remedy at each link. *(Analyze)*
3. A clinic next to a school has no segregation system. Explain the hospital risk-waste categories, the rules it is breaking, and who bears the greatest risk while they are broken. *(Evaluate)*
4. Argue the case that a city's investment in industrial effluent treatment pays back in water security, using the treatment stages and at least one figure from this unit. *(Evaluate)*
5. Design a plan outline for moving your school up the waste hierarchy: the first change, the second, who must agree to each, and a measurable end-of-term check. *(Create)*

Answers and marking guidance

MCQ answer key

1. b - about 97 of every 100 units are salt water in the seas.
2. b - security is reliable access to acceptable quality and quantity, not rainfall alone.
3. b - the study's samples exceeded WHO drinking-water guideline levels, including TDS.
4. b - nutrients over-fertilise water; the algal bloom and oxygen collapse follow.
5. b - secondary treatment is the biological stage; bacteria consume the organic load.
6. c - chromium is a metal and needs tertiary treatment; settling and dilution do not remove dissolved metals.
7. b - reduce, reuse, recycle, recover, dispose: prevention first, landfill last.
8. b - once streams mix, clean material is contaminated and recovery collapses.
9. b - clinical waste in the municipal stream puts infectious material in the open; the other three are failures too, but smaller.
10. b - ice caps, glaciers and groundwater hold most fresh water; rivers and lakes are a fraction of one per cent.

RRQ model answers and mark schemes

1. Model: a flood is the wrong water, contaminated and arriving at once, while drinking water needs a steady, clean, treated supply. Security is a match between need and usable supply, not total volume. **Marks (2):** 1 for the quality/timing point; 1 for the security definition applied.
2. Model: Indus river system, stressed and seasonal; glaciers and snowmelt, shrinking and shifting; groundwater, slowly recharged and often unsafe; rainfall, uneven and thin in the south. **Marks (4):** 1 per resource; 1 for at least two correct challenges.
3. Model: quality; the samples' physicochemical parameters, including TDS, exceeded WHO drinking guidelines, so the water is unusable even if the aquifer is not empty. **Marks (2):** 1 for quality; 1 for the finding as evidence.
4. Model: source, remedy: latrines or effluent treatment; pathway, remedy: protect the well head or treat the water; receptor, remedy: boil or chlorinate drinking water. **Marks (3):** 1 per link with its remedy.

5. Model: primary removes settleable solids; secondary removes dissolved organic matter biologically; tertiary removes metals, salts, nutrients or colour. **Marks (3):** 1 per stage.
6. Model: pathogens are invisible and cause disease; dissolved heavy metals or nitrate are invisible and toxic or persistent; clarity says nothing about either. **Marks (2):** 1 per pollutant type used correctly.
7. Model: the factory cuts its fresh-water draw and its discharge volume, and the river receives a smaller, cleaner load, so both the supply side and the pollution side of security improve. **Marks (2):** 1 for the factory benefit; 1 for the river benefit.
8. Model: waste never made needs no collection, sorting or processing, and recycling depends on clean separated material arriving, which reduce does not. **Marks (2):** 1 for the no-management-needed point; 1 for the dependency point.
9. Model: sharps, infectious waste, pharmaceutical waste (also pathological or chemical); sharps go in a sealed, puncture-proof container that cannot be reopened, handed to a licensed handler. **Marks (3):** 1 for three categories; 2 for the full sharps rule.
10. Model: argue for segregation, because the hierarchy's later steps, recycling, composting, safe disposal, all collapse on mixed waste, and trucks alone move the problem to the dumpsite, where the unit's leachate and methane evidence shows it does not disappear. **Marks (3):** 1 for a clear choice; 1 for hierarchy-based reasoning; 1 for use of the dumpsite evidence.

ERQ rubrics

1. **Analytic rubric.** *Distribution stated correctly (0-2) · Security reasoning: quality, quantity and timing separated (0-3, Analyze) · Two realistic actions, one household, one city (0-3, Apply) · Register suitable for a Class 8 pupil (0-2).* Total 10.
2. **Analytic rubric.** *Chain identified: source, pathway, receptor (0-2) · Pollutant types matched to the two discharges (0-3, Analyze) · One remedy at each link, matched to the pollutant (0-3, Apply) · Clarity (0-2).* Total 10.
3. **Analytic rubric.** *Risk-waste categories named (0-2) · Rules broken identified: segregation, storage, licensed transfer (0-3, Understand) · Risk-bearers named with reasoning: sweeper, picker, nurse, pupils (0-3, Evaluate) · Clarity (0-2).* Total 10.

4. **Analytic rubric.** *Treatment stages linked to the pollutants removed (0-3, Understand) · Water security argument: quantity and quality both addressed (0-3, Evaluate) · At least one unit figure used correctly, WHO sanitation or the dumpsite emissions (0-2, Apply) · Clarity (0-2). Total 10.*
5. **Analytic rubric.** *Changes ordered from reduce downward (0-3, Apply) · Feasibility: who agrees, what each costs (0-2) · Measurable end-of-term check defined (0-3, Analyze) · Clarity and register (0-2). Total 10.*

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